

Quantum Computing and Communications

Assignment-1

Ques-1 Non-orthogonal basis for $\mathbb{R}^2$: Consider the set of basis vectors $S = \left\{ |0\rangle, \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right\}$, where $|0\rangle$ and $|1\rangle$ are orthonormal vectors.

- 1) Write the co-ordinates of $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $|-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ in the new basis S .
- 2) If two vectors $|v_1\rangle$ and $|v_2\rangle$ have representations $\begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix}$ & $\begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix}$ in the new basis, work out the inner product $\langle v_1 | v_2 \rangle$. Hence, show that $\langle - | 1 \rangle$ remains the same in the new basis.

Ques-2 Consider a set of linearly dependent spanning vector in $\mathbb{R}^2 : \{|0\rangle, |1\rangle, |+\rangle\}$ $\left(|+\rangle \equiv \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$

- 1) Find out two representations (co-ordinates) of $|-\rangle$ and $|1\rangle$ in this set.
- 2) How many possible representations are there for any given vector? Justify your answer.

Ques-3 σ_X operator in the $\{|0\rangle, |1\rangle\}$ basis is given by its matrix representation $\sigma_X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Work out the matrix representation of σ_X in the basis $\{|+\rangle, |-\rangle\}$.

Ques-4 Find out the eigen values and eigen vectors of the following matrices and hence diagonalize them

- 1) $\sigma_X \equiv \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- 2) $\sigma_Y \equiv \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
- 3) $\sigma_Z \equiv \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- 4) $n_x \sigma_X + n_y \sigma_Y + n_z \sigma_Z (\equiv \vec{n} \cdot \vec{\sigma})$, where $n_x = \cos(\theta)$, $n_z = \sin(\theta)\cos(\phi)$, $n_y = \sin(\theta)\sin(\phi)$
- 5) $\sigma_X \otimes \sigma_X$
- 6) $\sigma_X \otimes \sigma_Y$
- 7) $|v_1\rangle\langle v_2|$ for some unit vectors $|v_1\rangle$ and $|v_2\rangle$

Note $\otimes$ denotes the Kronecker product. See https://en.wikipedia.org/wiki/Kronecker_product.

Ques-5 Consider two operators **A** and **B** in a vector space **V** and answer the following with proper justification.

- 1) If **A** and **B** are unitary, are $(\mathbf{A} + \mathbf{B})$, $\mathbf{AB}$, also unitary?
- 2) If **A** and **B** are Hermitian, are $\mathbf{AB}$, also Hermitian?
- 3) If **A** and **B** are Normal, are $(\mathbf{A} + \mathbf{B})$, $\mathbf{AB}$, also normal?